CP–7849

Sub. Code
11

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

TECHNICAL ENGLISH

(upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Give the antonyms of the following words :

   (a) love (b) hold

   (c) destroy

2. Write a note on the difference in spelling between British and American English.

3. Add either prefix or suffix of the following words :

   (a) important (b) posses

   (c) beauty

4. What are the different methods of interpreting graphics?

5. How do you guess the meaning of words from a context? Give some examples.

6. What are the aspects of a formal letter?

7. How do you adapt your ideas to audience?

8. In each of the following blanks write a verb in agreement with the subject.

   (a) Bread and butter ———————— (are/is) her favourite food.

   (b) A two thousand rupee ———————— (is/are) a good sum.

   (c) 'The Arabian Nights' ———————— (is/are) an interesting book.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Write an essay on body language.

Or

   (b) Define impersonal passive with examples.

10. (a) What is interpersonal communication?

Or

   (b) What are the different purposes of message in an organization?

11. (a) Write an essay on "Uniform Civil Code".

Or

   (b) Write a letter of complaint to the principal of a college about quality of food in the canteen.

12. (a) Write a letter to the editor of 'The Hindu' asking permission for the internship programme for the students of Technical English.

Or

   (b) Write a letter of application for the post of Junior Assistant in the Railway Department.

13. (a) Write an essay on the preparation for an Interview.

Or

(b) How does an oral presentation differ from a visual presentation?

CP–7850

Sub. Code
12

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

MATHEMATICS - I

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Find the rank of the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & 3 \\ 5 & -1 & 7 \end{bmatrix}$.

2. Show that a square matrix A and its transpose A^T have the same eigen values.

3. Direction ratios of a line are 3, 4, 12. What are its direction cosines.

4. Find the angle between the planes $2x + 4y - 6z = 11$ and $3x + 6y + 5z + 4 = 0$.

5. Find the radius of curvature for the curve $y = c\cos h\left(\frac{x}{c}\right)$ at $(0, c)$.

6. Find the evolute of the parabola $y^2 = 4ax$.

7. Find $\frac{dy}{dx}$ if $3x^2 + xy - y^2 + 4x - 2y + 1 = 0$.

8. Find the first and second order partial derivatives of $z = x^3 + y^3 - 3axy$.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Verify Cayley – Hamiltan theorem for the matrix $A = \begin{bmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{bmatrix}$.

Or

(b) Diagonalise the matrix $A = \begin{bmatrix} 2 & 2 & -7 \\ 2 & 1 & 2 \\ 0 & 1 & -3 \end{bmatrix}$ by similarity transformation.

10. (a) Find the equation of the plane passing through the point $(1, 2, -1)$ and perpendicular to the planes $x + y - 2z = 5$ and $3x - y + 4z = 12$.

Or

(b) Show that the lines $\frac{x-4}{2} = \frac{y-5}{3} = \frac{z-6}{4}$ and $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$ are coplanar, and find the equation of the plane in which they lie.

11. (a) Find the radius of curvature at the point $(a \cos^3 \theta, a \sin^3 \theta)$ on the curve $x^{2/3} + y^{2/3} = a^{2/3}$.

Or

(b) Find the envelope of a system of concentric ellipses with their axes along the co-ordinate axes and of constant area.

12. (a) If $u = f(x, y)$, where $x = r \cos \theta$, $y = r \sin \theta$, prove that $\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 = \left(\frac{\partial u}{\partial r}\right)^2 + \frac{1}{r^2}\left(\frac{\partial u}{\partial \theta}\right)^2$.

Or

(b) Find the Taylor's series expansion of $x^2y^2 + 2x^2y + 3xy^2$ in powers of $(x+2)$ and $(y-1)$ upto the third powers.

13. (a) Solve $(D^2 + 4)\, y = x^4 + \cos^2 x$.

Or

(b) Solve the equation $(1 - x^2)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + 2y = 2$, by the method of reduction of order.

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

PRODUCTION TECHNOLOGY

(Upto 2015 Batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Define step turning and taper turning.

2. Define single point cutting tool.

3. Explain Weldability.

4. List the sheet metal operations and explain any two operations.

5. Mention the types of lubricants and their uses.

6. Define various types of gears and its applications.

7. Mention different types of tolerances.

8. Explain the purpose of Vernier Caliper with its least count.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Explain various operations performed on lathe with suitable diagrams.

Or

(b) Explain various parts of lathe with neat sketch.

10. (a) What are the safety precautions to be taken for various processes? Explain.

Or

(b) Write short notes on :

(i) Welding.

(ii) Soldering

(iii) Brazing

(iv) Bronze welding.

11. (a) Explain various sheet metal operations used in sheet metal work.

Or

(b) Explain Basic principle of sheet metal work and what are all the tools in sheet metal work.

12. (a) Explain any two types of milling machine with suitable sketches.

Or

(b) Explain in detail about the numerical control machines and its advantages.

13. (a) Write short notes on :

(i) Micrometer

(ii) Vernier depth gauge with sketch.

Or

(b) Explain :

(i) Limits

(ii) Fits

(iii) Tolerances.

CP–7852

Sub. Code
14

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

ENGINEERING MECHANICS AND STRENGTH OF MATERIALS

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. State parallelogram law of forces.
2. Define statics and dynamics.
3. Define Equivalent forces.
4. Define simple harmonic motion.
5. Define friction.
6. Define perfect frame.
7. Define factor of safety.
8. Define couple lever arm principle.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Explain fundamental units and derived units.

Or

(b) Explain the principle of virtual work.

10. (a) Derive the equation of centre of gravity of the bodies.

Or

(b) Derive the equation of moment of inertia of bodies.

11. (a) Explain various laws of friction and co-efficient of friction.

Or

(b) Derive the equation of efficiency of screw jack.

12. (a) Explain various types of frames.

Or

(b) Explain resolution of forces using method of sections.

13. (a) Explain various types of forces.

Or

(b) Explain various types of beams.

———————

CP–7853

Sub. Code
21

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

MATHEMATICS — II

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Evaluate $\int_0^1 \int_0^2 xy^2 \, dy \, dx$.

2. Evaluate $\int_0^2 \int_0^1 \int_0^2 xyz \, dx \, dy \, dz$.

3. If $\bar{r}$ is the position vector joining the origin of a co-ordinate system and any point $p(x, y, z)$, prove that grad $\left(\frac{1}{r}\right) = \frac{-\bar{r}}{r^3}$.

4. Find the value of a if $\bar{F} = (axy - z^3)i + (a-2)x^3 j + (1-a) xz^2 \bar{k}$ is irrotational.

5. Examine the analyticity of the function $f(z) = z^2$.

6. Evaluate $\int \frac{z \, dz}{z-2}$ where C is the circle $|z| = 1$.

7. Find $L(t \sin 3t \cos 2t)$.

8. Find $L\left[\frac{S}{(S+2)^2 + 1}\right]$.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Evaluate $\iint (x^2 + y^2)\, dx\, dy$ over the region for which x, y each ≥ 0 and $x + y \leq 1$.

Or

(b) Evaluate $\iiint xyz\, dx\, dy\, dz$ taken through the positive octant of the sphere $x^2 + y^2 + z^2 = a^2$.

10. (a) Evaluate $\iint \overline{A} \cdot \overline{n}\, ds$ if $\overline{A} = zi + x\overline{j} - y^2 z\overline{k}$ and S is the surface of the cylinder $x^2 + y^2 = 1$ in the first octant be the planes $z = 0$ and $z = 2$.

Or

(b) Verify the divergence theorem for $\overline{A} = (x + y)\, i + x\overline{j} + z\overline{k}$ taken over the region V of the cube bounded by the planes $x = 0$, $x = 1$, $y = 0$, $y = 1$, $z = 0$, $z = 1$.

11. (a) Show that the function $u = e^x (\cos y - \sin y)$ is harmonic and find its harmonic conjugates.

Or

(b) Find the analytic function $f(z) = a + iv$ if $u + v = \dfrac{x}{x^2 + y^2}$ and $f(1) = 1$.

12. (a) Obtain Taylor's series to represent the function $\frac{z^2 - 1}{(z+2)\,(z+3)}$ in the region $|z| < 1$.

Or

(b) Find the residue of $\frac{z}{(z-1)^4\,(z-2)\,(z-3)}$.

13. (a) Find $L^{-1}\left[\frac{1}{S(S+1)\,(S+2)}\right]$.

Or

(b) Solve $\frac{d^2y}{dt^2} + 2\frac{dy}{dt} = 8t$ with $y = 0 = \frac{dy}{dt}$ at $t = 0$.

CP–7854

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22

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

FLUID MECHANICS

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. State Pascal's law with an example.

2. Mention various properties of fluid.

3. Define centre of pressure.

4. Discuss the types of fluid flow.

5. Define Bernoulli's principle.

6. Define rate of flow.

7. Define continuity equation.

8. Define turbulent flow.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Explain various properties of fluids.

Or

(b) Explain the principle of working of mechanical gauges.

10. (a) Explain the metacentric height of submerged and floating bodies.

Or

(b) Explain the principle of working of accumulator with a sketch.

11. (a) Explain streamlines, stream functions with examples.

Or

(b) Explain source flow and sink flow with diagrams.

12. (a) Deduce Bernoulli's equation from Euler's equation from the 1st principle.

Or

(b) Derive the equation of discharge through orificemeter.

13. (a) Explain the laws of fluid friction and laminar flow.

Or

(b) Explain various forces exerted by a fluid on a stationary body with an equation.

CP–7855

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23

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

THERMODYNAMICS

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Explain about thermodynamic equilibrium.

2. State law of conservation of momentum.

3. Why Cp is greater Cv?

4. Explain gas compression.

5. State Dalton's law for partial pressure.

6. Define heating value of fuel.

7. Explain isothermal efficiency.

8. Differentiate closed and open cycle of turbine.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) State first law of thermodynamics and obtain Mayer's relation of specific heat capacities.

Or

(b) Write physical signtigence of entropy and also calculate the increase in entropy when 1 gram of ice at –10°C is converted into steam.

(Given : SHC of ice 2100 J/Kg/K. Latent heat of ice : $3.35 \times 10^5 J/Kg$. Latent heat steam : $26 \times 10^6 J/Kg$ SHC of water : 4200 J/Kg/K.

10. (a) Deduce laws of ideal gas on the basis of kinetic theory of gases.

Or

(b) Describe Carnot cycle with neat diagram and also obtain its efficiency equation.

11. (a) Discuss the following.

(i) Properties of mixture of gases.

(ii) Enthalpy and specific heats of gas mixtures.

Or

(b) Derive an expression for the determination of Minimum air required for combustion.

12. (a) Discuss in detail about reciprogating air compressor. With neat diagram.

Or

(b) Describe the construction and working of Aixal flow compressor with neat diagram.

13. (a) Describe the working of open cycle gas turbine with neat diagram.

Or

(b) Explain in detail the following.

(i) Jet propulsion

(ii) Rocket propulsion.

———————

CP–7856

Sub. Code
24

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

BASIC ELECTRICITY AND ELECTRONICS

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. State and explain Ohm's law.
2. What is a DC generator?
3. State how single phase induction motor works.
4. Mention some applications of function generators.
5. How half wave rectifier works?
6. What is leakage factor?
7. Give the types of transistors.
8. Explain power loss in a transformer.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) (i) State and explain Kirchoffs law.

(ii) State and explain norton's theorem.

Or

(b) Discuss nodal voltage analysis, mechanical analysis.

10. (a) What are the different types of generators? Explain the characteristics of generators.

Or

(b) Describe working of transformer and explain the different types of single phase transformer.

11. (a) Explain briefly slip torque characteristics of cage and wound rotor.

Or

(b) Briefly explain the construction and principle of operations of induction motor.

12. (a) What is UJT? Describe its construction and characteristics mention its application.

Or

(b) With necessary circuit diagram explain the operation of multimeter, voltmeter and ammeter.

13. (a) Explain the construction and working of a full wave rectifier with neat circuit diagram. Calculate its rectification efficiency.

Or

(b) Write notes on:

(i) Voltage regulator

(ii) SMPS

(iii) Transformer utility factor.

CP–7857

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31

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

MECHANICS OF FLIGHT

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Define ISA and its reference points.

2. Define Bernoulli's Principles.

3. What is venture effect?

4. Define shock wave.

5. Define take off run and its significance.

6. Define turbulent flow and its significance.

7. Write three axes of aircraft and its relative motion?

8. List out the names of control surfaces used in transport aircraft.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Explain about the seven layers of atmosphere and its significance related to aircraft performance.

Or

(b) Write short notes on (i) Define altitude and its various types (ii) Physical conditions of atmosphere.

10. (a) Describe with neat sketch, describe the various types of wind tunnel.

Or

(b) Write short notes on (i) Autorotation (ii) manuverability (iii) various forces acting on helicopter.

11. (a) Write the types of supersonic airfoils and its character.

Or

(b) Write short notes on (i) Streamline flow (ii) Laminar flow (iii) Turbulent flow.

12. (a) Briefly describe about the concept of lift and its importance.

Or

(b) What is drag and its various types?

13. (a) Explain the operation and the effects of aileron for roll motion.

Or

(b) List out and explain the dynamic problems of the aircraft stability.

CP–7858

Sub. Code
32

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

AIRCRAFT MATERIALS

(Upto 2015 Batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Write the character of nickel alloy?
2. How will you distinguish between the Ferrous and Non-Ferrous metals?
3. Define creep.
4. What is the role of NDT?
5. Write the significance of Anodising?
6. Write the few names of Doping agents?
7. Write the significance of DACRON fabric?
8. Write the importance of Graphite fibers?

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Explain the titanium alloy purpose and their applications?

Or

(b) Describe about the aluminium alloy and its purposes.

10. (a) Write the purpose and procedure for Rockwell Hardness test?

Or

(b) Describe about the Dye Penetrant Method?

11. (a) Explain about the Metal spraying?

Or

(b) Explain about the Heat Treatment process for Non-Ferrous metals.

12. (a) Write short notes on (i) Blind Rivet (ii) Standard Cherry lock Rivet.

Or

(b) List out the properties of thermosetting resin.

13. (a) List out the application of composite materials and its advantages.

Or

(b) Describe about the core material of sandwich construction.

———————

CP–7859

Sub. Code
33

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

AIRCRAFT STRUCTURES

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. What is sandwich construction?

2. What is fuselage and its types?

3. What is the purpose of rudder?

4. What is called Spoiler and its importance?

5. What is tandem landing gear?

6. Write the types of wheel assembly used in aircraft?

7. Write the role of electronic load cells?

8. Define Dihedral angle.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Explain the semi-monocoque construction and its advantages.

Or

(b) Explain any one types of Honeycomb construction.

10. (a) What is flap and its various types with neat sketch?

Or

(b) Describe about the fly-by-wire system.

11. (a) Explain in details about the power boost brake system of an aircraft.

Or

(b) List out and explain the Retraction Test procedure.

12. (a) How will you compute the C.G. location of aircraft?

Or

(b) List out the weighing procedure of an aircraft.

13. (a) Briefly explain about the Rigging of an aircraft.

Or

(b) Write short notes on :

(i) Symmetric check

(ii) Surface alignment check.

———————

CP–7860

Sub. Code
34

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

AIRCRAFT SYSTEMS

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Write the principles of Pascal's Law.

2. Write the types of seals used in aircraft hydraulic system?

3. What are the purposes of pneumatic system?

4. Write the significance of NRV in the pneumatic system?

5. What is the role of pressurisation system?

6. Write the significance of dump valve in pressurisation system?

7. What is ACM?

8. Define anti-icing and de-icing.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Describe about the types of fluid and its characteristics?

Or

(b) Sketch and explain the hydraulic system layout of Boeing 747?

10. (a) Explain about the Pneumatic system of an small executive aircraft?

Or

(b) Write short notes on

(i) Pressure regulator (ii) Pressure relief valve.

11. (a) Briefly describe about the Pressure Controller in pressurisation system?

Or

(b) Explain the various components of air-conditioning system.

12. (a) Sketch and explain the Thermal anti-icing system of an small aircraft.

Or

(b) What is wind shield wiper and its construction and operation details?

13. (a) Explain the various types of fuels used in piston and gas turbine engines?

Or

(b) Write short notes on

(i) types of fuel tanks

(ii) fuel contamination and its checks.

———————

CP–7861

Sub. Code
41

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

PISTON ENGINE AND PROPELLER

(Upto 2015 Batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Differentiate Boyle's law and Charles's law.
2. Describe Otto cycle.
3. Write a note on propeller shafts.
4. Write a note on waste gate of turbo charger.
5. List out the advantages of fuel injection system.
6. Explain the laws of evaporation.
7. How magnetos may be classified?
8. What is geometric pitch of propeller?

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) What is four strokes five event cycles and explain the valve position in an exhaust stroke?

Or

(b) Explain the various factors which affect the performance of a piston engine.

10. (a) Describe the simple induction system with neat diagram.

Or

(b) Draw an external type supercharger system and explain.

11. (a) Discuss about main metering system of a float carburetor.

Or

(b) Discuss about aircraft oil filtering systems and various oil filters.

12. (a) Describe about the engine operating controls.

Or

(b) Name various auxiliary ignition units and explain about any two.

13. (a) Explain about various propeller clearances.

Or

(b) Describe the various forces acting on the propeller in flight.

———————

CP–7862

Sub. Code
42

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Sciences

GAS TURBINE ENGINE

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Describe propulsive efficiency.
2. Describe bypass ratio.
3. Write short notes on 'Magnetic Chip Detector'.
4. Define Potential energy and kinetic energy.
5. What is the purpose of the dump valve in fuel system?
6. Define compressor stall.
7. Describe specific fuel consumption.
8. What is the purpose of scavenge oil pump.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Explain in detail the principles of Jet propulsion.

Or

(b) Describe briefly the events of the Braytons cycle with volume-pressure and Temperature pressure diagram.

10. (a) What are the various types of compressor are used in Gas turbine Engine? Explain its operation with advantages and disadvantages.

Or

(b) What are the various types of combustion chambers used in Gas turbine Engine? Explain them by neat sketches with advantages and disadvantages of each.

11. (a) Briefly explain the various types of forces acting on a propeller.

Or

(b) Explain the operation of a Turboprop engine.

12. (a) Explain Full Authority Digital Electronic Control (FADEC) System.

Or

(b) Describe various types of jet fuel and explain the characteristics in detail.

13. (a) Explain the working principle of Air turbine starter.

Or

(b) Describe main components of ignition system of a Gas turbine engine.

———————

CP–7863

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43

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017.

Aeronautical Science

AIRCRAFT INSTRUMENTS

(Upto 2015 Batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Define ISA.

2. Differentiate scale length and scale marking

3. What are square law and its compensation in A.S.I?

4. What are thermocouple and its application?

5. What do you understand by oxygen plumbing?

6. What do you understand by Error's of deviation and Co-efficient in compass?

7. Define the three 'Q' codes used for altimeter pressure setting.

8. What do you understand about pitot and static pressure?

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) What is the colour range marking on ASI and its indications?

Or

(b) What is Head Up Display and explain its construction and operation?

10. (a) With a schematic diagram state the construction and operation of continuous flow type oxygen system?

Or

(b) Describe about barometric pressure ceiling of altimeter.

11. (a) What are the various types of thermocouples used on piston and jet engine? Explain its construction and operation.

Or

(b) Explain the operation of a Vertical Speed Indicator (VSI) when aircraft goes from a level flight into a climb attitude.

12. (a) Briefly explain the operating procedure of capacitance type fuel quantity indicator.

Or

(b) Describe about the EPR instrument.

13. (a) Briefly discuss about the remote reading compass swinging procedure.

Or

(b) Explain about the compass errors.

CP–7864

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44

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

AIRCRAFT ELECTRICAL SYSTEM

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Define voltage, current and Resistance.

2. Define atom, Elements and compound.

3. What do you mean by armature reaction and its methods of elimination?

4. Name various types of circuit protection devices.

5. Write the uses of interpole.

6. List down different types of lights and its purpose in aircraft.

7. What is Hall Effect Probe?

8. What are the rules for systematic trouble shooting?

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Explain the direction of current flow.

Or

(b) Write notes on voltage potential difference and EMF (Electro Motive Force).

10. (a) Explain the theory of Lead Acid battery.

Or

(b) Briefly explain about the voltage regulator.

11. (a) Explain about the equalizing Circuit.

Or

(b) Explain the construction details of Armature assembly in DC Generator.

12. (a) Explain about turbine engine auto ignition circuit.

Or

(b) Explain briefly about parallel and split bus bar system.

13. (a) Write in detail about CRO and its function.

Or

(b) Explain the trouble shooting of light circuits and logic flow chart.

———————

CP–7865

Sub. Code
51

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Sciences

AIRCRAFT RULES AND AIRWORTHINESS REGULATION – I

(Upto – 2015 Batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Write briefly about the preservation of log books.

2. What are the different log books to be maintained for aircraft registered in India?

3. Explain

   (a) Aircraft imported by Air

   (b) Change of ownership

   (c) Issue of duplicate certificate of registration.

4. Define :

   (a) Certificate of approval

   (b) Active Mandatory Modification

   (c) Problematic use of substances.

5. Define:

   (a) Unusable fuel

   (b) Fuel dip stick

   (c) Fuelling and Fuelling zone.

6. Define :

   (a) Aircraft

   (b) Aerial works

   (c) Aerodrome.

7. Write down the procedure for changing the ownership of an Aircraft.

8. Define

   (a) Psycho active substances

   (b) Aircraft goods

   (c) Accountable Manager.

Part B $(5 \times 12 = 60)$

Answer **all** questions.

9. (a) Write down the details about Journey log book.

Or

   (b) Write down the procedure and steps to be followed for registration of aircraft.

10. (a) Write down the location of nationality on common make and registration marks on

   (i) Lighter than an aircraft

   (ii) Heavier than an aircraft.

Or

   (b) Write down the documents to be submitted for validation of type certificate of an aircraft.

11. (a) Explain

    (i) Positioning of fuelling equipment near aircraft

    (ii) Manning of fuelling vehicles

    (iii) Approval of fuel samples and starting of fuelling.

Or

(b) What are the special precautions to be taken in the fuelling zone?

12. (a) Write the safety precautions against fire hazard.

Or

(b) Write down the steps to be followed against safety precautions against static electricity discharge, Bonding and Earthling.

13. (a) Discuss the steps to be followed while servicing and maintenance of aircraft during fuelling.

Or

(b) Write down the precautionary measures to be observed while fuelling with passengers aboard and during embarkation and disembarkation.

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

AIRCRAFT MAINTENANCE

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Describe major maintenance.

2. What is malfunction of Defect report?

3. Describe inter-granular corrosion.

4. What are the effects of out of balance of control surfaces?

5. What are the effects of excessive tension of control cable?

6. Write the functions of landing gear.

7. Differentiate Thermoplastics and Thermosetting plastics?

8. State the safety precautions to be observed while using pneumatic gun.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Write notes on Maintenance Records.

Or

(b) Explain preflight and post flight inspection?

10. (a) Describe the principles involved in protection against corrosion.

Or

(b) Describe the special checks to be carried out for heavy landing.

11. (a) State the inspection procedure for typical control system.

Or

(b) Describe the purpose of Balancing or rebalancing of flying control surfaces.

12. (a) Describe the inspection of Aircraft Wheels.

Or

(b) Explain the procedure for tire mounting and demounting on a wheel assembly.

13. (a) How will you carry out repair a hole on plastic structure?

Or

(b) Explain the following forming operations :

(i) Bumbing.

(ii) Stretching.

(iii) Shrinking.

(iv) Crimping.

CP–7867

Sub. Code
53

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

AEROENGINE MAINTENANCE

(Upto 2015 Batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Explain the basic categories of On-wing maintenance.
2. List out the purpose of local etching.
3. Define Pre-heating of an engine.
4. What is blade life in propeller?
5. What are the types of propeller shafts?
6. Describe 'hung start' of a jet engine.
7. How will you assess engine performance?
8. What is the function of carburetor?

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Explain in detail about general overhaul procedure of gas turbine engine.

Or

(b) What do you understand by NDT checks? Briefly explain about magnetic particle testing.

10. (a) Enumerate wooden propeller defects and repair procedure.

Or

(b) List out the general inspection and repair of metal propellers.

11. (a) Explain the procedure of ignition system operational checks during Ground run.

Or

(b) Briefly explain the reciprocating engine, shutdown and post stopping procedure.

12. (a) Explain about full spray nozzle testing procedure.

Or

(b) Describe about over speed and overheat inspection procedure of Gas turbine engine.

13. (a) Describe the procedure for full throttle operation checks and assessment of engine performance.

Or

(b) How will you assess the engine performance in terms of engine pressure ratio parameter?

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CP–7868

Sub. Code
54

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

AIRCRAFT COMMUNICATION AND NAVIGATION SYSTEM

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. What is TACAN?
2. What are the functions of Antenna coupler?
3. What are the procedures for testing communication radio?
4. What are the functions of transmitter?
5. Describe Course Deviation Indicator.
6. What is GPWS?
7. What is radio telephone?
8. What are the functions of Radar?

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Draw a block diagram of superhetrodyne receiver and explain.

Or

(b) Describe Selcal Decoder with a help of a diagram.

10. (a) Describe about LORAN.

Or

(b) Describe VOR receiver.

11. (a) Explain the operation of Instrument Landing system.

Or

(b) Explain inertial Navigation System.

12. (a) Explain the operation of ELT.

Or

(b) Explain Distance Measuring Equipment.

13. (a) Describe Traffic Alert and Collision Avoidance system.

Or

(b) Describe about colour weather Radar.

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CP–7869

Sub. Code
61

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

AIRCRAFT RULES AND AIRWORTHINESS REGULATIONS — II

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Define the following terms.
   (a) Aerial work
   (b) Commercial air transport operation
   (c) Glider
   (d) General aviation operation.

2. What are the instruments and equipments required for navigator?

3. What are the items to be equipeed with on all seaplanes for all flights?

4. Write down the items equipped with all aircrafts when operated on extended flights over water.

5. Define the following.
   (a) Flight recorder
   (b) Commercial operation
   (c) General aviation.

6. Write short notes on calibration of FDR system.

7. Define

   (a) Cockpit Audio Recording System (CARS)

   (b) Cockpit Voice Recorder (CVR)

   (c) Ground Proximity Warming System (GPWS)

8. Under what circumstances the GPWS should give warning automatically?

Part B $(5 \times 12 = 60)$

Answer **all** questions.

9. (a) What are the instruments, equipments and flight documents to be carried in addition to the minimum equipment for the issuance of a certificate of airworthiness?

Or

   (b) What are the instruments and equipments required for flying club and aerial work operations?

10. (a) Write down the instruments and equipments installed on all aircraft when operated at night?

Or

   (b) Write the instruments equipped in all aircraft operated in accordance with the Instrument Flight Rule (IFR).

11. (a) What are the types of flight data recorder? Explain each.

Or

   (b) Explain aeroplane-commercial air transports which are fitted with FDR's.

12. (a) Write briefly about helicopter-commercial air transport and general aviation which are fitted with different types of FDR.

Or

(b) Explain inspection of flight recorder system.

13. (a) What are the general requirements of Cockpit Voice Recorder (CVR) and Cockpit Audio Recording System (CARS)?

Or

(b) Write the instruments equipped with all aircraft when operated as VFR flights.

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62

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

INDUSTRIAL MANAGEMENT

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. What is operational planning?

2. What are the functions of management?

3. Distinguish between authority and power.

4. What do you mean by manpower?

5. Write the process of decision making.

6. How do you manage records?

7. What do you mean by work study?

8. What are the objectives of ergonomics?

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) What do you mean by management? Discuss the functions of management.

Or

(b) Explain in detail about management by objective.

10. (a) Discuss about the elements of organizational structure.

Or

(b) Write in detail about the bases of power.

11. (a) Explain the importance and methods of motivation.

Or

(b) Discuss in detail about the role and duties of a supervisor.

12. (a) What do you mean by time management? Explain the techniques of time management?

Or

(b) Explain the process involved in production management.

13. (a) What is meant by inventory? Discuss the techniques in inventory control.

Or

(b) Explain in detail about man machine system.

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CP–7871

Sub. Code
63

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

AIRPORT AND AIR TRAFFIC SERVICES

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. What is greenfield airport?
2. What is LCN?
3. Name the different aircraft parking configuration with neat sketch.
4. Define Holding Apron and Terminal Apron.
5. What is an X-ray unit? Why it is installed in an airport?
6. List out the functions of briefing centre in an airport.
7. What are the goods prohibited for carriage by air?
8. Write the objectives of air traffic control.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Describe the principles of airport layout with neat sketch.

Or

(b) Explain DGCA and its functions.

10. (a) Explain runway markings and taxiway markings with neat sketch.

Or

(b) Describe in detail about Ramp safety requirements.

11. (a) Explain flight plan and its inscriptions.

Or

(b) Briefly describe the use of NDB, VOR, DME and ILS.

12. (a) Explain Airport lightings with neat sketch.

Or

(b) Narrate the Airport security methods and techniques.

13. (a) What are the divisions of ATC? Explain in detail about the ATC services.

Or

(b) Explain the role of meteorology and importance of communication in aviation industry.

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CP–7872

Sub. Code
64

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

GROUND HANDLING, SAFETY AND SUPPORT EQUIPMENT

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Part A $(5 \times 3 = 15)$

Answer any **five** questions.

1. Explain NFPA standard #10 regarding marking of fire extinguishers.

2. Describe six methods of fire extinguishing operation.

3. Describe standard taxi light signals.

4. How verities of aircraft not aligning with tie down anchor points are secured?

5. Describe the procedure to be followed for spilled oil and grease.

6. What are the advantages of single point fueling?

7. Describe pre-oiling equipment.

8. How the GPU is parked near the aircraft?

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Describe the flight line safety.

Or

(b) Briefly explain the aircraft fire extinguishers and name the extinguishing agents not recommended for aircraft use with reasons.

10. (a) Describe the precautions to be followed against windstorm damage.

Or

(b) Describe the securing of

(i) Sea plane and aircraft on SKIS

(ii) Multi engine aircraft.

11. (a) State maintenance safety on compressed air system.

Or

(b) (i) State the power cord protection.

(ii) State the precautions for aircraft tire mounting.

12. (a) Describe the duties of refueling crew.

Or

(b) Explain the four principal contaminants in aviation fuel which reduces the quality.

13. (a) Describe the operating procedure and maintenance of Hydraulic Test Rig.

Or

(b) Describe the tow bar and its servicing.